

Challenge: Super SSH

Category: General Skills

Description: The challenge required connecting to a remote server using SSH (Secure Shell), a protocol for securely logging into and controlling a remote machine over a network, in order to retrieve a flag stored on that system.

Approach: The challenge provided the exact connection details needed: a username (ctfplayer), a hostname (titan.picoctf.net), a specific port (51460, rather than SSH's default port 22), and a password. I connected using the ssh command, specifying the custom port with the -p flag, since the server wasn't listening on the standard SSH port. On first connection, I was prompted to confirm the server's authenticity (its "fingerprint") — a normal SSH security check the first time you connect to any new server — which I accepted by typing yes. After entering the provided password, I was logged into the remote system's command line. From there, I used ls to view the files present and cat to read the contents of the flag file directly.

Difficulty Faced: As my first CTF challenge, I wasn't yet familiar with SSH's command syntax or with the idea that a server could run SSH on a non-default port. I had to learn what each part of the connection command meant — the username format, the -p flag for a custom port, and why accepting a server's fingerprint on first connection is a normal, expected step. Once I understood the command structure, connecting and retrieving the flag was straightforward.

Tools used: ssh (OpenSSH client)

Command used: ssh ctf-player@titan.picoctf.net -p 51460

Flag: picoCTF{s3cur3_c0nn3ct10n_07a987ac}

What I learned: SSH is the secure counterpart to insecure remote-access protocols like telnet or rexec, which I encountered earlier in my vulnerability assessment — those transmit credentials in plaintext, while SSH encrypts the entire session. This challenge also reinforced that SSH doesn't always run on its default port (22); servers can be configured to listen on any port, which is exactly why the connection command needed the explicit -p flag.

Evidence:

```
(kali@kali)-[~]
$ ssh ctf-player@titan.picoctf.net -p 51460
The authenticity of host '[titan.picoctf.net]:51460 ([3.139.174.234]:51460)' can't be established.
ED25519 key fingerprint is: SHA256:4S9EbTSSRZm32I+cdM5TyzthpQryv5kudRP9PIKT7XQ
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[titan.picoctf.net]:51460' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
ctf-player@titan.picoctf.net's password:
Welcome ctf-player, here's your flag: picoCTF{s3cur3_c0nn3ct10n_07a987ac}
Connection to titan.picoctf.net closed.
```